Contribution, Credits & Capacity

The complete glossary of Trusty's contribution economy · engine v1 · July 2026

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1 · Contribution

Contribution measures how much a person has made the Trust Network *smarter* — more accurate, more complete, or more useful for future decisions. It is determined by **impact, not effort**: a single recommendation that repeatedly leads to verified good outcomes creates more contribution than a hundred generic ones.

> **The one question.** Every action is judged by: *did this leave the knowledge graph objectively better than before?* If yes, it is a candidate contribution. If no, it is ordinary app activity and earns nothing.

Contribution is **never stored as a score**. It is recomputed live from evidence (recommendations, outcomes, approved corrections) every time it is read — which means the whole history can always be replayed under a future, improved model.

2 · What counts — the kind registry

Only actions in a closed registry earn value. Each has a category and a base value (the worth of one fully-accepted event):

KIND	CATEGORY	BASE VALUE	WHAT IT MEANS
`recommendation_created`	Creation	2	You published a recommendation — new knowledge for future decisions.
`entity_added`	Creation	2	You added a missing place or specialist to the network.
`photo_added`	Creation	1	Your photo (approved) helps others recognize a real place.
`dish_added`	Creation	1	You documented a dish (approved) — menus become more complete.
`outcome_recorded`	Validation	1	You reported how a recommendation actually worked out — real-world evidence.
`recommendation_validated`	Validation	**3**	Someone *else* reported a positive outcome on *your* recommendation. The strongest signal — your knowledge genuinely helped a real person. Counted once per independent person, never per repeat.
`entity_correction`	Improvement	2	Your suggested edit to a place/specialist was approved by moderation.
`entity_maintenance`	Maintenance	1.5	You kept records accurate (hours, closures, availability). Grows in value as the graph matures.

Validation deliberately outweighs creation (3 > 2): confirming that knowledge is true reduces uncertainty more than adding unconfirmed knowledge. Adding a future contribution type = one registry entry + an evidence source — no engine redesign.

3 · What never counts

> **Structurally worth zero.** These aren't just down-weighted — they are absent from the registry, so even if recorded they aggregate to exactly 0:

- Logging in / session time
- Clicks, searches, views
- Profile visits
- Follower or listener counts
- Subscription status
- Money spent — **ever**
- Sending invitations (alone)
- Number of recommendations *alone* (see burst damping)
- Comments
- Being popular
- Anything an AI "grants" — AI may suggest, only validated graph improvement counts

Guiding principle: *users are rewarded for making the network smarter — not for making it busier.*

4 · The lifecycle

Every contribution event moves through a lifecycle, and its value depends on where it stands:

STATUS	MULTIPLIER	MEANING
recorded	× 0.25	Submitted, not yet validated — e.g. a suggested edit awaiting moderation. Earns a small provisional value.
validated	× 1.0	Confirmed by evidence — an outcome recorded, an independent verification.
accepted	× 1.0	Live in the graph — an approved edit, a published recommendation.
rejected	× **0**	Declined by moderation. **Stays in history forever** — but counts for nothing. History is preserved; interpretation evolves.

> **Nothing is rewarded just for being submitted.** The platform first establishes that the graph genuinely improved. That's why a pending suggestion is worth a quarter of an approved one.

5 · The score & its mechanics

The **Contribution Score** summarizes accumulated impact. It's explainable, config-driven, and computed as:

```
event value  =  base value × lifecycle multiplier × burst damping
core  =  Lifetime × 0.7  +  Momentum × 0.3
score  =  core × Consistency × Diversity
```

Burst damping — quality over quantity

Repeating the *same kind* of action within one day: each repeat is worth **0.7×** the previous one (the n-th same-day event is worth 0.7^{n-1}). Fifty recommendations pasted in one afternoon are worth a fraction of the same fifty spread over weeks — flooding doesn't pay, and a weekend spammer can never outrank a steady monthly contributor.

6 · Lifetime & Momentum

Lifetime (weight 0.7)

The sum of all your event values, **never decayed**. Contribution does not evaporate because time passes: a great recommendation from two years ago still counts fully here. If you go inactive, your history waits for you (recovery, not erasure).

Momentum (weight 0.3)

The same events, but each weighted by recency with a **90-day half-life**: an event from 90 days ago counts half here, from 180 days a quarter, and so on. Momentum is the "actively engaged right now" signal.

> **Balance by design.** A fresh contribution earns ≈ its full value (0.7 + 0.3×1.0). A very old one still earns 0.7×. Momentum never erases history; history never completely drowns out recent value.

7 · Consistency & Diversity

Consistency (up to × 1.5)

Counts your distinct **active months** (months with at least one valued contribution). One month is the baseline (×1.0); the multiplier rises smoothly and saturates around +50%. Someone contributing monthly for two years ends up meaningfully ahead of someone with one busy weekend — *contribution behaves like professional credibility: earned by repeatedly creating value.*

Diversity (up to × 1.15)

A small bonus for improving *different parts* of the graph — creation, validation, improvement, maintenance. Four categories touched ≈ +15%. A diverse contributor builds more resilient knowledge than someone repeating one action forever. It's a signal, deliberately small — never a farming target.

8 · Maturity stages

Human-readable levels derived from the score (thresholds configurable, presentation-independent):

STAGE	MIN SCORE		STAGE	MIN SCORE
New Contributor	0		Trusted Contributor	90
Occasional Contributor	5		Knowledge Builder	180
Active Contributor	15		Community Expert	350
Reliable Contributor	40			

9 · Listen & Listen Capacity

Listen is Trusty's deliberate replacement for "Follow". Listening to someone means: "*I want this person's future recommendations to become part of my recommendation network.*" It is directional, requires the other person's acceptance, and — frozen guarantee — **never creates trust, never raises reputation, never changes any score**. It only makes their knowledge *visible* to you.

Listen Capacity is how many people you may actively listen to. It is a limited, earned resource — the network grows deliberately, not virally:

```
capacity = base (5) + ⌊score / 5⌋ (contribution) + ⌊weeks of membership / 4⌋
(tenure, max 10) + active boosts · hard cap 50
```

- **Base** — every account starts with 5 slots after onboarding.
- **Contribution** — the primary growth path: one slot per 5 score points.
- **Tenure** — one slot per 4 weeks of membership, capped at 10: capacity also grows slowly just for remaining a member, independent of contribution.
- **Boosts** — temporary extra slots bought with credits (below); they sit on top of the normal ceiling because they expire.

- Removing a Listen connection frees its slot instantly. Capacity is never permanently lost.
- **Capacity is never purchasable with money.** There is deliberately no payment input anywhere in the formula.

10 · Credits & Boosts

Credits are controlled access to platform capabilities. They are **not** currency, **not** transferable, **not** trust, and **not** contribution — and they can never influence trust, reputation, or how recommendations are ranked.

```
balance = earned + ledger  where  earned = ⌊score / 2⌋ (derived live, never stored),  ledger = grants − spends (append-only, auditable)
```

- **Earned credits** derive automatically from your validated contribution: 1 credit per 2 score points.
- **Grants** — an admin can grant credits; every grant records who granted, when, and why.
- **Spending (v1): the capacity boost** — 3 credits buy **+1 Listen slot for 30 days** (at most 3 boosts active at once). Future spends: advanced AI analysis, deeper graph exploration, premium workflows.
- Every credit movement lives in an append-only ledger — fully auditable, nothing is ever deleted.

> **Credits vs. Capacity.** Capacity controls *how much of the network you may connect with*. Credits control *which advanced capabilities you may consume*. Neither affects trust; neither affects ranking.

11 · The four independent pillars

PILLAR	ANSWERS	SOURCE	FEEDS
Trust	"Can this recommendation be relied on?"	Directed, topic-scoped trust edges + verified outcomes; computed per request, never stored	Trust search & recommendations
Reputation	"How reliable is this provider in this topic?"	Outcome history (Bayesian, time-decayed, consensus-by-independence)	Ranking (one factor of several)
Relationship Strength	"How close are these two people?"	Edge priors + decayed real interactions	Personalization
Contribution	"How much has this person helped the network learn?"	The evidence-derived engine in this document	Listen capacity, credits, future rewards — **never ranking, never trust**

> **Independence is structural.** Contribution never creates trust; trust never guarantees contribution. Listening never creates either. Payment influences none of them. These aren't policies enforced by review — the code paths simply don't connect.

12 · Explainability & auditability

- **Every score decomposes into factors** — "Your recommendations produced 2 independently verified outcomes", "Your 3 approved corrections improved place profiles". Users never see unexplained points.
- **Every event is auditable** — evidence lives in append-only stores (outcome log, moderation queue, credit ledger). Rejected work stays visible in history at zero value.
- **Everything is replayable** — since scores are derived, a future improved model can rescore the complete history. The admin "what-if simulator" is exactly this: recompute everyone under different knobs, compare, commit only if better.

13 · Tuning knobs (env vars)

VARIABLE	DEFAULT	CONTROLS
`CONTRIB_HALF_LIFE_DAYS`	90	Momentum half-life
`CONTRIB_BURST_DECAY`	0.7	Same-day repeat damping
`CONTRIB_W_LIFETIME` / `CONTRIB_W_MOMENTUM`	0.7 / 0.3	Lifetime vs momentum blend
`LISTEN_BASE_SLOTS`	5	Starting capacity
`LISTEN_CONTRIBUTION_PER_SLOT`	5	Score points per earned slot
`LISTEN_WEEKS_PER_SLOT` / `LISTEN_MAX_TENURE_SLOTS`	4 / 10	Tenure growth rate & cap (0 disables)
`LISTEN_MAX_SLOTS`	50	Hard capacity ceiling
`CREDITS_POINTS_PER_CREDIT`	2	Score points per earned credit
`LISTEN_BOOST_COST` / `LISTEN_BOOST_DAYS` / `LISTEN_BOOST_SLOTS` / `LISTEN_BOOST_MAX_ACTIVE`	3 / 30 / 1 / 3	The capacity boost

The admin panel's what-if simulator previews any of these without changing the live system; making a tuning permanent means setting the env var and restarting.

14 · A worked example

Maria joins Trusty in March.

- **March:** she publishes a recommendation for her immigration lawyer (`recommendation_created`, +2) and reports how a restaurant recommendation worked out (`outcome_recorded`, +1). Score ≈ 3 → *New Contributor*, 5 slots, 1 credit.

- **April:** a colleague hires Maria's lawyer and reports a great outcome → `recommendation_validated`, +3 — her strongest event, earned by someone *else's* verified experience. She also suggests a correction to a clinic's hours; while pending it's worth 0.5 (2 × 0.25), after approval the full 2.
- **May:** her score is ≈ 8, boosted a little by two active months (consistency) and three categories touched (diversity) → *Occasional Contributor*, 6 slots (5 + ⌊8/5⌋), 4 credits.
- **July:** four months of membership add a tenure slot (⌊17 weeks / 4⌋ = 4 → capped growth). She spends 3 credits on a boost before a conference: +1 slot for 30 days. Capacity: 5 base + 1 contribution + 4 tenure + 1 boost = 11.
- **What never moved her score:** the 400 searches she ran, her 12 profile views, the 3 friends she invited who never contributed, and the month she paid for a future premium plan.

Trusty · Contribution Engine v1 glossary · generated July 2026 · sources: GitHub issues #20–#23 and `packages/graph-core/src/contribution.ts`, `listen.ts`, `services/graph-service/src/credits.ts`. All numbers shown are the current defaults; every one of them is configurable.